PRACTICAL NO 6 : FACE DE-AGING / AGING

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import torch
from PIL import Image
from diffusers import AutoPipelineForImage2Image

#model
model_id="Stable-diffusion-v1-5/stable-diffusion-vl-5"

#select device
device="cuda" if torch.cuda.is_available()else"cpu"

print("Using device:",device)

#load image-to-image pipeline
if device == "cuda":
   pipe=AutoPipelineForImage2Image.from_pretrained(
      model_id,
      torch_dtype=torch.float16,
      use_safetensors=True
   )
else:
   pipe=AutoPipelineForImage2Image.from_pretrained(
      model_id,
      torch_dtype=torch.float32,
      use_safetensors=True
   )


pipe =pipe.to(device)

#load input face
image=Image.open("face.jpg").convert("RGB")
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#resize image
imageimage.resize((512,512))


print("\nChoose an Option:")
print("1. Face Aging")
print("2. Face De-Aging")


choice = input("Enter choice (1 or 2): ")
if choice == "1":
    prompt = (
        "realistic aged version of the same person,"
        "older adult face, natural wrinkles, gray hair,"
        "realistic skin texture,preserve facial identity,"
        "photorealistic"
        )
elif choice == "2":
    prompt = (
        "realistic younger version of the same person,"
        "younger adult face, smooth natural skin,"
        "youthful appearance,preserve facial identity,"
        "photorealistic"
       )

else:
    print("Invalid choice!")
    exit()

print("\nGenerating image... Please wait.")

#generate image
result =pipe(
    prompt=prompt,
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    image=image,
    strength=0.45,
    guidance_scale=7.5,
    num_inference_steps=30
    ).images[0]

#save output
result.save("face_transformed.png")

print("\nDone!")
print("Output saved as : face tramsformed.png")
```

Output :

Original Image



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Using device: cuda
Loading pipeline components...: 100% 7/7 [00:01<00:00, 4.50it/s]
Loading weights: 100% 196/196 [00:00<00:00, 1515.16it/s]
Loading weights: 100% 396/396 [00:00<00:00, 1135.77it/s]

Choose an Option:
1. Face Aging
2. Face De-Aging
Enter choice (1 or 2):  1

Generating image... Please wait.
100% 13/13 [00:02<00:00, 6.53it/s]

Done!
Output saved as : face_transformed.png
```

1.Face Ageing



2.Anti Face Ageing

